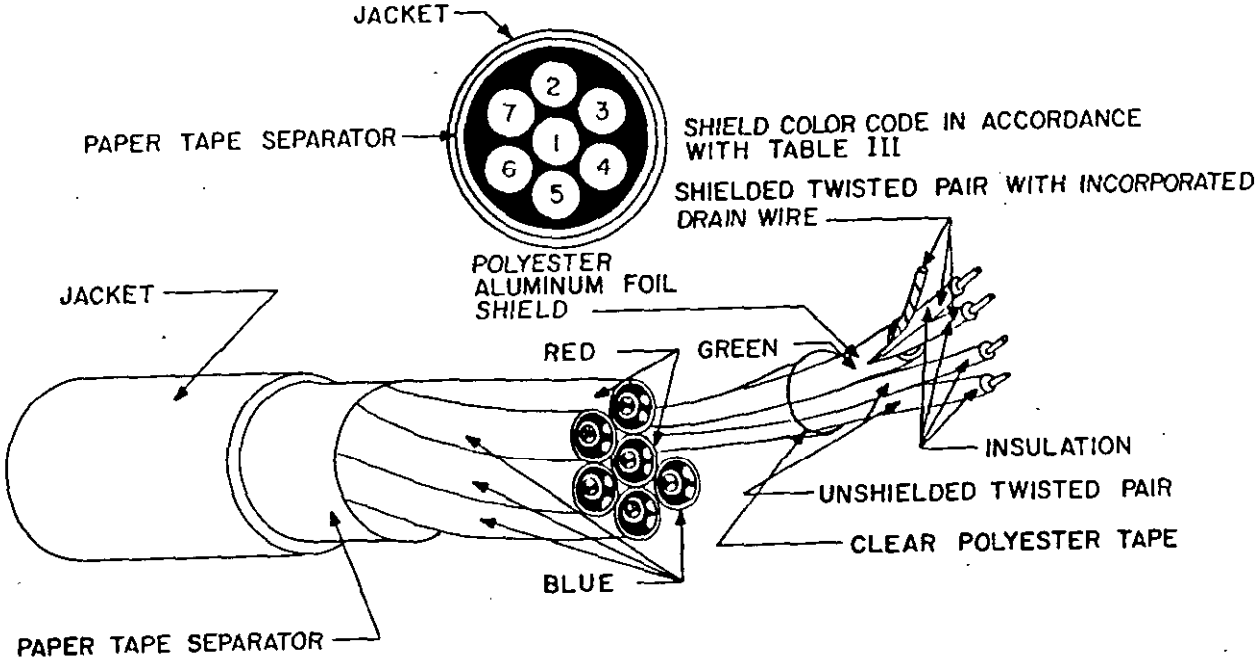
MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, SEVEN QUADS CONSISTING OF TWO SHIELDED PAIRS OF 22 (7 X 30) AWG

[NOTE: NOT FOR AEROSPACE USE.]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.



14 PAIRS CONFIGURED INTO 7 QUADS

FIGURE 1. Cable illustration.

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Engineering information: (See table I).

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	34 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	67 pF/ft
Continuous working voltage	200 V rms maximum between conductors
Nominal capacitance between adjacent shields	150 pF/ft
Continuous working voltage between adjacent shields	50 volts maximum
Shield dc resistance	20.5 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of quads	Seven
Conductor type and wire size	Stranded tinned copper 22 AWG
Conductor stranding	7 X 30 AWG
Conductor insulation	Polypropylene
Conductor insulation thickness	0.008 inch nominal thickness
Drain wire type and size	Stranded tinned copper 24 AWG
Drain wire stranding	7 X 32 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.040 inch
Finished cable diameter	0.457 inch maximum outside diameter
Cable style (UL)	Not applicable
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	1.3 twists per foot ±10%

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The shield color code indicates the color of the polyester portion of the aluminum foil shield involved with the shielded pair. Each quad contains one shielded pair and one unshielded pair.

The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1 and table III).

TABLE III. Shield color code.

Quad number	Shield color
Center core	
1st quad	Red
Outer layer	
2nd quad	Red
3rd quad	Green
4th quad	Blue - 1
5th quad	Blue - 2
6th quad	Blue - 3
7th quad	Blue - 4

REQUIREMENTS:

Design and construction: (See tables II and III).

Quad configuration. Each quad contains two pairs. An individually shielded pair is cabled with a second pair and both pairs are covered with a clear polyester tape.

Shield location and orientation. The individually shielded pair is covered with a polyester aluminum shield located on the outer circumference of the pair with aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact. A polyester tape is applied over the individually shielded pair and the second pair.

Drain wire location. Each drain wire is to be spirally located within the individually shielded pair shield. It is to be in continuous contact with the aluminum surface of the shield throughout the cable. This construction contains one drain wire per quad.

Polyester tape. A plastic barrier tape is to be applied over each quad.

Polypropylene tape. A separator tape shall be placed between the jacket and the cabled pairs. The tape shall be approximately 1.25 inches wide, applied as a spiral wrap with a minimum of 3.75 inches overlap. The polypropylene separator tape shall have a left hand lay.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 1581 VW-1 flame test requirements.

Cabling. Each quad is to be cabled on a common axis to reduce the overall diameter.

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Conductor color code. The first pair constitutes the individually shielded pair component of the quad.

1st pair (shielded pair)		2nd pair (under overall shield)	
1st conductor:	Red	1st conductor:	Green
2nd conductor:	Black	2nd conductor:	White

Shield effectiveness evaluation test connections. An individual quad will be tested as if it were one pair. Remove the jacket and remove an individual quad from the cable. Install the quad in the test fixture. Test the shielded pair with the unshielded conductors floating within the quad. Care must be taken in order to preserve the integrity of the construction during testing.

Shield integrity test.* The following list of responses refers to the pair covered by an individual pair shield and located within the overall shield.

Frequency	Response **
100 kHz	114 dB below reference
500 kHz	102 dB below reference
1 MHz	95 dB below reference
5 MHz	79 dB below reference
10 MHz	72 dB below reference
15 MHz	67 dB below reference
20 MHz	63 dB below reference
25 MHz	60 dB below reference
30 MHz	57 dB below reference
100 MHz	44 dB below reference

Fixture resonance will occur between 40 and 70 MHz.

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable shall have a durometer hardness of 85 ±5.

Insulation resistance between shields: 100 megohms/1,000 feet (minimum).

Part or Identifying Number (PIN): The PIN shall be M49285/29.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1114-30)

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